Supplementary Information for "Exact Selective Locality Reduction for Resource-Constrained Quantum Approximate Optimization"

Jingcong Liu, Yiding Su, Ruojun Zhang, Hafiz Muhammad Waseem, Bo Chen, and Carsten Maple

Abstract

This document contains the proofs, algorithmic details, benchmark generation rules, software configuration, provenance records, and extended numerical results that support the main text. Section, figure, table, and equation numbers carry the prefix S.

CONTENTS

S1 Proofs 4
- S1-A Sharp One-Step Threshold (Proof of Proposition 1) . . . 4
- S1-B Sequential Compositional Exactness (Proof of Theorem 1) . . . 4
- S1-C Product Grammar and Target Locality . . . 4
- S1-D Auxiliary-Reuse Structure and Pair Cover . . . 4
- S1-E Validity of the Selection Certificate . . . 4

S2 Selection Algorithms 4
- S2-A Reference Compilation Model . . . 4
- S2-B Fixed Device-Level Protocol . . . 4
- S2-C Fibre Diagnostics in Closed Form . . . 4
- S2-D Candidate Evaluation, Beam Search, and Branch and Bound . . . 4
- S2-E Stopping Rules and Returned Status . . . 4

S3 Warm-Start Circuits and Diagnostics 4
- S3-A Matched Mixer and Cost-Layer Factorisation . . . 4
- S3-B Original-Objective and Consistency Diagnostics . . . 4
- S3-C Alternative Auxiliary Initialisations . . . 4
- S3-D Warm-Start Policies and Null Control . . . 4

S4 Benchmark Construction 4
- S4-A Max-3SAT Expansion and Worked Instance . . . 4
- S4-B Cubic Spin-Glass Conversion . . . 4
- S4-C Generation, Screening, and Split Assignment . . . 4
- S4-D Relaxation and Optimum-Face Selection . . . 4
- S4-E Fully Reduced and Matched-Random References . . . 4

S5 Software, Compiler Configuration, and Seeds 4
- S5-A Software Versions and Transpiler Settings . . . 4
- S5-B Coupling Maps . . . 4
- S5-C Optimisation Hyperparameters and Seeds . . . 4
- S5-D Noise Model . . . 4

Corresponding author: Hafiz Muhammad Waseem (e-mail: HM.Waseem@warwick.ac.uk).

S6 Provenance and Reproducibility Records 4
S6-A Archive Identities and Retained Records 4
S6-B Warm-Start Coordinate Correction 4
S6-C Selector Repair and Merged Result Files 4
S6-D Origin of Each Reported Result 4

S7 Extended Numerical Results 4
S7-A Complete Exactness Counts 4
S7-B Selector Validation, Certification, and Sensitivity 4
S7-C Marginal Resource Levels 4
S7-D Additional Coupling Maps 4
S7-E Zero-Layer Accounting and Secondary QAOA Diagnostics 4
S7-F Warm-Start Diagnostics 4
S7-G Descriptive Regime Counts 4
S7-H Limited Noise Study 4

References 4

S1. Proofs

A. Sharp One-Step Threshold (Proof of Proposition 1)
B. Sequential Compositional Exactness (Proof of Theorem 1)
C. Product Grammar and Target Locality
D. Auxiliary-Reuse Structure and Pair Cover
E. Validity of the Selection Certificate

S2. Selection Algorithms

A. Reference Compilation Model
B. Fixed Device-Level Protocol
C. Fibre Diagnostics in Closed Form
D. Candidate Evaluation, Beam Search, and Branch and Bound
E. Stopping Rules and Returned Status

S3. Warm-Start Circuits and Diagnostics

A. Matched Mixer and Cost-Layer Factorisation
B. Original-Objective and Consistency Diagnostics
C. Alternative Auxiliary Initialisations
D. Warm-Start Policies and Null Control

S4. Benchmark Construction

A. Max-3SAT Expansion and Worked Instance
B. Cubic Spin-Glass Conversion
C. Generation, Screening, and Split Assignment
D. Relaxation and Optimum-Face Selection
E. Fully Reduced and Matched-Random References

S5. Software, Compiler Configuration, and Seeds

A. Software Versions and Transpiler Settings
B. Coupling Maps
C. Optimisation Hyperparameters and Seeds
D. Noise Model

S6. Provenance and Reproducibility Records

A. Archive Identities and Retained Records
B. Warm-Start Coordinate Correction
C. Selector Repair and Merged Result Files
D. Origin of Each Reported Result

S7. Extended Numerical Results

A. Complete Exactness Counts
B. Selector Validation, Certification, and Sensitivity
C. Marginal Resource Levels
D. Additional Coupling Maps
E. Zero-Layer Accounting and Secondary QAOA Diagnostics
F. Warm-Start Diagnostics
G. Descriptive Regime Counts
H. Limited Noise Study

References

[1] Endre Boros and Peter L. Hammer. “Pseudo-boolean optimization”. Discrete Applied Mathematics **123**, 155–225 (2002).

[2] Stuart Hadfield. "On the representation of Boolean and real functions as Hamiltonians for quantum computing". ACM Transactions on Quantum Computing **2**, Article 18, 1–21 (2021). arXiv:1804.09130.

[3] Edward Farhi, Jeffrey Goldstone, and Sam Gutmann. "A quantum approximate optimization algorithm" (2014). arXiv:1411.4028.

[4] Leo Zhou, Sheng-Tao Wang, Soonwon Choi, Hannes Pichler, and Mikhail D. Lukin. "Quantum Approximate Optimization Algorithm: Performance, Mechanism, and Implementation on Near-Term Devices". Physical Review X **10**, 021067 (2020).

[5] Adam Glos, Aleksandra Krawiec, and Zoltán Zimborás. "Space-efficient binary optimization for variational quantum computing". npj Quantum Information **8**, 39 (2022). arXiv:2009.07309.

[6] Colin Campbell and Edward Dahl. "QAOA of the highest order". In 2022 IEEE 19th International Conference on Software Architecture Companion (ICSA-C), pages 141–146 (2022). doi:10.1109/ICSA-C54293.2022.00035. arXiv:2111.12754.

[7] Jonas Stein, Farbod Chamanian, Maximilian Zorn, Jonas Nüßlein, Sebastian Zielinski, Michael Kölle, and Claudia Linnhoff-Popien. "Evidence that PUBO outperforms QUBO when solving continuous optimization problems with the QAOA". In Proceedings of the Genetic and Evolutionary Computation Conference Companion. Pages 2254–2262 (2023). arXiv:2305.03390.

[8] Kristina Bell, Adam Lowe, Emily Coles, Nathanael Ridgway, and Gillian Marshall. "A comparison of quadratic and higher-order representations for QAOA" (2025). arXiv:2509.20127.

[9] Sabina Drăgoi, Alberto Baiardi, and Daniel J. Egger. "Approximate quadratization of high-order Hamiltonians for combinatorial quantum optimization". Physical Review Research **8**, 023159 (2026). arXiv:2505.04700.

[10] Zoé Verchère, Sourour Elloumi, and Andrea Simonetto. "Optimizing variational circuits for higher-order binary optimization". In 2023 IEEE International Conference on Quantum Computing and Engineering (QCE), volume 1, pages 19–25 (2023). doi:10.1109/QCE57702.2023.00011. arXiv:2307.16756.

[11] Johannes Weidenfeller, Lucia C. Valor, Julien Gacon, Caroline Tornow, Luciano Bello, Stefan Woerner, and Daniel J. Egger. "Scaling of the quantum approximate optimization algorithm on superconducting qubit based hardware". Quantum **6**, 870 (2022). arXiv:2202.03459.

[12] Matthew P. Harrigan et al. "Quantum approximate optimization of non-planar graph problems on a planar superconducting processor". Nature Physics **17**, 332–336 (2021).

[13] I. G. Rosenberg. "Reduction of bivalent maximization to the quadratic case". Cahiers du Centre d'Études de Recherche Opérationnelle **17**, 71–74 (1975).

[14] Endre Boros and Aritanan Gruber. "On quadratization of pseudo-boolean functions" (2014). arXiv:1404.6538.

[15] Nike Dattani. "Quadratization in discrete optimization and quantum mechanics" (2019). arXiv:1901.04405.

[16] Daniel Freedman and Petros Drineas. "Energy minimization via graph cuts: Settling what is possible". In 2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05), volume 2, pages 939–946 (2005). doi:10.1109/CVPR.2005.143.

[17] Hiroshi Ishikawa. "Transformation of general binary MRF minimization to the first-order case". IEEE Transactions on Pattern Analysis and Machine Intelligence **33**, 1234–1249 (2011).

[18] Martin Anthony, Endre Boros, Yves Crama, and Aritanan Gruber. "Quadratic reformulations of nonlinear binary optimization problems". Mathematical Programming **162**, 115–144 (2017).

[19] Martin Anthony, Endre Boros, Yves Crama, and Aritanan Gruber. "Quadratization of symmetric pseudo-Boolean functions". Discrete Applied Mathematics **203**, 1–12 (2016).

[20] Endre Boros, Yves Crama, and Elisabeth Rodríguez-Heck. "Compact quadratizations for pseudo-Boolean functions". Journal of Combinatorial Optimization **39**, 687–707 (2020).

[21] Amit Verma and Mark Lewis. "Optimal quadratic reformulations of fourth degree pseudo-Boolean functions". Optimization Letters **14**, 1557–1569 (2020).

[22] Amit Verma, Mark Lewis, and Gary Kochenberger. "Efficient QUBO transformation for higher degree pseudo Boolean functions" (2021). arXiv:2107.11695.

[23] Yves Crama, Sourour Elloumi, Amélie Lambert, and Elisabeth Rodríguez-Heck. "Quadratization and convexification in polynomial binary optimization". Journal of Combinatorial Optimization **50**, Article 28 (2025).

[24] Rebekah Herrman, Lorna Treffert, James Ostrowski, Phillip C. Lotshaw, Travis S. Humble, and George Siopsis. "Globally optimizing QAOA circuit depth for constrained optimization problems". Algorithms **14**, 294 (2021). arXiv:2108.03281.

[25] Lukas Schmidbauer, Karen Wintersperger, Elisabeth Lobe, and Wolfgang Mauerer. "Polynomial reduction methods and their impact on QAOA circuits". In 2024 IEEE International Conference on Quantum Software (QSW), pages 35–45 (2024). doi:10.1109/QSW62656.2024.00018. arXiv:2406.08889.

[26] Lukas Schmidbauer, Elisabeth Lobe, Ina Schaefer, and Wolfgang Mauerer. "It's quick to be square: Fast quadratisation for quantum toolchains". ACM Transactions on Quantum Computing **7**, Article 18, 1–46 (2026). arXiv:2411.19934.

[27] Damian Rovara, Lukas Burgholzer, and Robert Wille. "Quantum hardware-efficient selection of auxiliary variables for QUBO formulations". In 2026 Design, Automation & Test in Europe Conference (DATE), pages 1–7 (2026). doi:10.23919/DATE69613.2026.11539407. arXiv:2511.19613.

[28] Daniel J. Egger, Jakub Mareček, and Stefan Woerner. "Warm-starting quantum optimization". Quantum **5**, 479 (2021).

[29] Reuben Tate, Majid Farhadi, Creston Herold, Greg Mohler, and Swati Gupta. "Bridging classical and quantum with SDP initialized warm-starts for QAOA". ACM Transactions on Quantum Computing **4**, Article 9, 1–39 (2023).

[30] Reuben Tate, Jai Moondra, Bryan Gard, Greg Mohler, and Swati Gupta. "Warm-started QAOA with custom mixers provably converges and computationally beats Goemans–Williamson's Max-Cut at low circuit depths". Quantum **7**, 1121 (2023).

[31] Zhihui Wang, Stuart Hadfield, Zhang Jiang, and Eleanor G. Rieffel. "Quantum approximate optimization algorithm for MaxCut: A fermionic view". Physical Review A **97**, 022304 (2018).

[32] Edward Farhi, David Gamarnik, and Sam Gutmann. "The Quantum Approximate Optimization Algorithm Needs to See the Whole Graph: Worst Case Examples". Preprint at arXiv:2005.08747 (2020).

[33] V. Akshay, H. Philathong, M. E. S. Morales, and J. D. Biamonte. "Reachability Deficits in Quantum Approximate Optimization". Physical Review Letters **124**, 090504 (2020).

[34] Madelyn Cain, Edward Farhi, Sam Gutmann, Daniel Ranard, and Eugene Tang. "The QAOA gets stuck starting from a good classical string". Preprint at arXiv:2207.05089v2 (2023).

[35] A. H. Land and A. G. Doig. "An Automatic Method of Solving Discrete Programming Problems". Econometrica **28**, 497–520 (1960).

[36] E. L. Lawler and D. E. Wood. "Branch-and-Bound Methods: A Survey". Operations Research **14**, 699–719 (1966).

[37] Stuart Hadfield, Zhihui Wang, Bryan O'Gorman, Eleanor G. Rieffel, Davide Venturelli, and Rupak Biswas. "From the Quantum Approximate Optimization Algorithm to a Quantum Alternating Operator Ansatz". Algorithms **12**, 34 (2019).

[38] M. J. D. Powell. "A Direct Search Optimization Method That Models the Objective and Constraint Functions by Linear Interpolation". In Advances in Optimization and Numerical Analysis, pages 51–67 (1994).

[39] Student. "The Probable Error of a Mean". Biometrika **6**, 1–25 (1908).

[40] Andrew Lucas. "Ising formulations of many NP problems". Frontiers in Physics **2**, 5 (2014).

[41] Fred Glover, Gary Kochenberger, and Yu Du. "A Tutorial on Formulating and Using QUBO Models". Preprint at arXiv:1811.11538 (2018).

[42] Alexander Cowtan, Silas Dilkes, Ross Duncan, Will Simmons, and Seyon Sivarajah. "Phase Gadget Synthesis for Shallow Circuits". Electronic Proceedings in Theoretical Computer Science **318**, 213–228 (2020).

[43] Alexander Cowtan, Will Simmons, and Ross Duncan. "A Generic Compilation Strategy for the Unitary Coupled Cluster Ansatz". Preprint at arXiv:2007.10515 (2020).

[44] Prakash Murali, Jonathan M. Baker, Ali Javadi-Abhari, Frederic T. Chong, and Margaret Martonosi. "Noise-Adaptive Compiler Mappings for Noisy Intermediate-Scale Quantum Computers". Proceedings of the Twenty-Fourth International Conference on Architectural Support for Programming Languages and Operating Systems, 1015–1029 (2019).

[45] Kalyanmoy Deb, Amrit Pratap, Sameer Agarwal, and T. Meyarivan. "A fast and elitist multiobjective genetic algorithm: NSGA-II". IEEE Transactions on Evolutionary Computation **6**, 182–197 (2002).

[46] Joao Basso, David Gamarnik, Song Mei, and Leo Zhou. "Performance and limitations of the QAOA at constant levels on large sparse hypergraphs and spin glass models". In 2022 IEEE 63rd Annual Symposium on Foundations of Computer Science (FOCS), pages 335–343 (2022). doi:10.1109/FOCS54457.2022.00039. arXiv:2204.10306.

[47] Elijah Pelofske, Andreas Bärtschi, and Stephan Eidenbenz. "Short-depth QAOA circuits and quantum annealing on higher-order Ising models". npj Quantum Information **10**, 30 (2024).

[48] Elijah Pelofske, Andreas Bärtschi, Lukasz Cincio, John Golden, and Stephan Eidenbenz. "Scaling whole-chip QAOA for higher-order Ising spin glass models on heavy-hex graphs". npj Quantum Information **10**, 109 (2024). arXiv:2312.00997.

[49] Ali Javadi-Abhari et al. "Quantum computing with Qiskit". Preprint at arXiv:2405.08810 (2024).

[50] Pauli Virtanen et al. "SciPy 1.0: fundamental algorithms for scientific computing in Python". Nature Methods **17**, 261–272 (2020).

[51] Gushu Li, Yufei Ding, and Yuan Xie. "Tackling the Qubit Mapping Problem for NISQ-Era Quantum Devices". Proceedings of the Twenty-Fourth International Conference on Architectural Support for Programming Languages and Operating Systems, 1001–1014 (2019).

[52] Gavin E. Crooks. "Performance of the Quantum Approximate Optimization Algorithm on the Maximum Cut Problem". Preprint at arXiv:1811.08419 (2018).

[53] Stefan H. Sack and Maksym Serbyn. "Quantum annealing initialization of the quantum approximate optimization algorithm". Quantum **5**, 491 (2021).

[54] Q. Huangfu and J. A. J. Hall. "Parallelizing the dual revised simplex method". Mathematical Programming Computation **10**, 119–142 (2018).

[55] Jeffrey Marshall, Filip Wudarski, Stuart Hadfield, and Tad Hogg. "Characterizing local noise in QAOA circuits". IOP SciNotes **1**, 025208 (2020).

[56] Samson Wang et al. "Noise-induced barren plateaus in variational quantum algorithms". Nature Communications **12**, 6961 (2021).

[57] Daniel Stilck França and Raul García-Patrón. "Limitations of optimization algorithms on noisy quantum devices". Nature Physics **17**, 1221–1227 (2021).

[58] Sebastian Nagies, Kevin T. Geier, Javed Akram, Dimitrios Bantounas, Michael Johanning, and Philipp Hauke. "Boosting quantum annealing performance through direct polynomial unconstrained binary optimization". Quantum Science and Technology **10**, 035008 (2025). arXiv:2412.04398.

[59] Jonas Nüßlein, Sebastian Zielinski, Thomas Gabor, Claudia Linnhoff-Popien, and Sebastian Feld. "Solving (Max) 3-SAT via quadratic unconstrained binary optimization" (2023). arXiv:2302.03536.

[60] Jonathan Wurtz and Peter Love. "MaxCut quantum approximate optimization algorithm performance guarantees for $p > 1$". Physical Review A **103**, 042612 (2021).

[61] Joao Basso, Edward Farhi, Kunal Marwaha, Benjamin Villalonga, and Leo Zhou. "The Quantum Approximate Optimization Algorithm at High Depth for MaxCut on Large-Girth Regular Graphs and the Sherrington-Kirkpatrick Model". Leibniz International Proceedings in Informatics (TQC 2022) **232**, 7:1–7:21 (2022).

[62] Hanif D. Sherali and Warren P. Adams. "A Hierarchy of Relaxations between the Continuous and Convex Hull Representations for Zero-One Programming Problems". SIAM Journal on Discrete Mathematics **3**, 411–430 (1990).

[63] Garth P. McCormick. "Computability of global solutions to factorable nonconvex programs: Part I — Convex underestimating problems". Mathematical Programming **10**, 147–175 (1976).

[64] Frederik Koch, Shahram Panahiyan, Rick Mukherjee, Joseph Doetsch, et al. "Resource-efficient quantum optimization via higher-order encoding". EPJ Quantum Technology (2026). arXiv:2511.17545.

[65] Puya Mirkarimi, David C. Hoyle, Ross Williams, and Nicholas Chancellor. "Quantum optimization with linear Ising penalty functions for customer data science". New Journal of Physics **26**, 103005 (2024).

[66] P. V. Sriluckshmy, Vicente Pina-Canelles, Mario Ponce, Manuel G. Algaba, Fedor Šimkovic, and Martin Leib. "Optimal, hardware native decomposition of parameterized multi-qubit Pauli gates". Quantum Science and Technology **8**, 045029 (2023).

[67] Matthew Amy, Parsiad Azimzadeh, and Michele Mosca. "On the CNOT-complexity of CNOT-PHASE circuits". Quantum Science and Technology **4**, 015002 (2018).

[68] Adam Holmes, Sonika Johri, Gian Giacomo Guerreschi, James S. Clarke, and A. Y. Matsuura. "Impact of qubit connectivity on quantum algorithm performance". Quantum Science and Technology **5**, 025009 (2020).

[69] Alwin Zulehner, Alexandru Paler, and Robert Wille. "An efficient methodology for mapping quantum circuits to the IBM QX architectures". IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems **38**, 1226–1236 (2019).
[70] Benjamin Tan, Marc-Antoine Lemonde, Supanut Thanasilp, Jirawat Tangpanitanon, and Dimitris G. Angelakis. "Qubit-efficient encoding schemes for binary optimisation problems". Quantum **5**, 454 (2021).
[71] Rebekah Herrman, James Ostrowski, Travis S. Humble, and George Siopsis. "Lower bounds on circuit depth of the quantum approximate optimization algorithm". Quantum Information Processing **20**, 59 (2021).
[72] Amira Abbas et al. "Challenges and opportunities in quantum optimization". Nature Reviews Physics **6**, 718–735 (2024).
[73] Mayowa Ayodele. "Penalty weights in QUBO formulations: Permutation problems". In Evolutionary Computation in Combinatorial Optimization (EvoCOP 2022), Lecture Notes in Computer Science **13222**, pages 159–174 (2022).